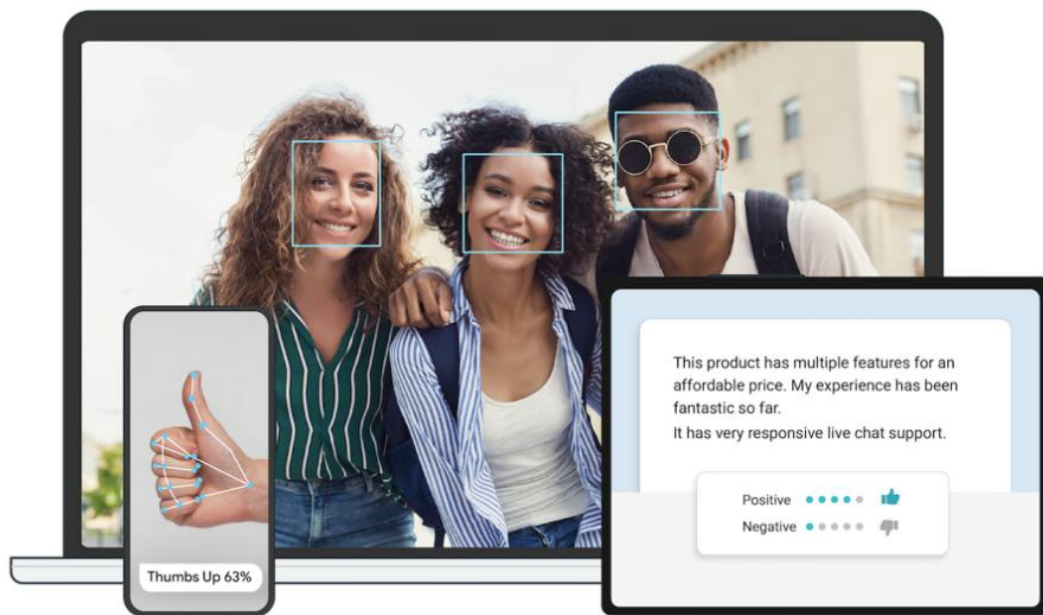


MediaPipe Introduction

1. MediaPipe Introduction

1.1 MediaPipe Description

MediaPipe is an open-source framework of multi-media machine learning models. Cross-platform MediaPipe can run on mobile devices, workspace and servers, as well as support mobile GPU acceleration. It is also compatible with TensorFlow and TF Lite Inference Engine, and all kinds of TensorFlow and TF Lite models can be applied on it. Besides, MediaPipe supports GPU acceleration of mobile and embedded platform.



1.2 MediaPipe Pros and Cons

1.2.1 MediaPipe Pros

- 1) MediaPipe supports various platforms and languages, including iOS, Android, C++, Python, JAVAScript, Coral, etc.
- 2) Swift running. Models can run in real-time.
- 3) Models and codes are with high reuse rate.

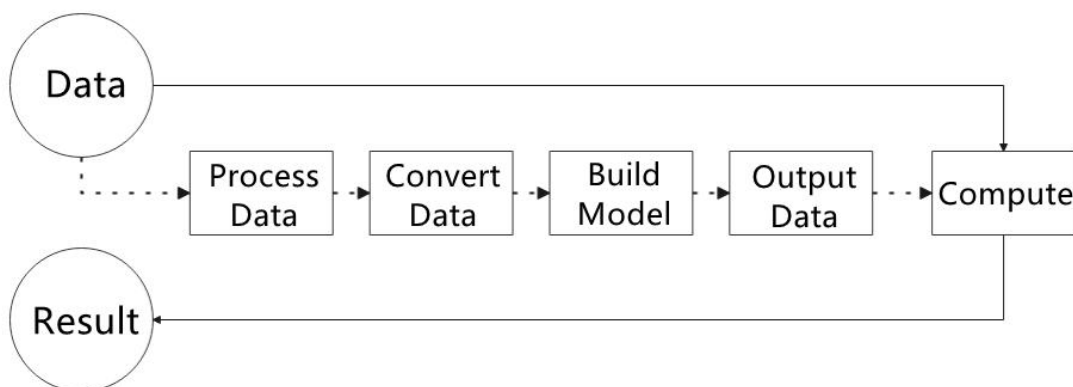
1.2.2 MediaPipe Cons

- 1) For mobile devices, MediaPipe will occupy 10M or above.
- 2) As it greatly depends on Tensorflow, you need to alter large amount of codes if you want to change it to other machine learning frameworks, which is not friendly to machine learning developer.
- 3) It adopts static image which can improve efficiency, but make it difficult to find out the errors.

1.3 How to use MediaPipe

The figure below shows how to use MediaPipe. The solid line represents the part to be coded, and the dotted line indicates the part not to be coded.

MediaPipe can offer the result and the function realization framework quickly.



1.3.1 Dependency

MediaPipe utilizes OpenCV to process video, and uses FFMPEG to process audio data. Furthermore, it incorporates other essential dependencies, including OpenGL/Metal, Tensorflow, and Eigen.

For seamless usage of MediaPipe, we suggest gaining a basic understanding of OpenCV. To delve into OpenCV, you can find detailed information in "4. OpenCV Computer Vision Lesson".

1.3.2 MediaPipe Solutions

Solutions is based on the open-source pre-constructed sample of TensorFlow or TFLite. MediaPipe Solutions is built upon a framework, which provides 16 Solutions, including face detection, Face Mesh, iris, hand, posture, human body and so on.

1.4 MediaPipe Learning Resources

MediaPipe website: <https://developers.google.com/mediapipe>

MediaPipe Wiki: <http://i.bnu.edu.cn/wiki/index.php?title=Mediapipe>

MediaPipe github: <https://github.com/google/mediapipe>

Dlib website: <http://dlib.net/>

dlib github: <https://github.com/davisking/dlib>